Electric Vehicles and Battery Technology

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Introduction to Electric Vehicles

An electric vehicle (EV) is a vehicle that is powered wholly or partly by an electric motor using energy stored in a battery, instead of an internal combustion engine burning petrol or diesel. Battery electric vehicles (BEVs) run only on electricity and have zero tailpipe emissions. Hybrid electric vehicles (HEVs) combine an engine with a small battery and motor, plug-in hybrids (PHEVs) can also be charged from a socket, and fuel cell electric vehicles (FCEVs) generate electricity from hydrogen. EVs include two-wheelers, three-wheelers, cars, buses and trucks. They matter because road transport is a major source of air pollution and greenhouse gases, and because India imports most of its crude oil.

History

Electric cars are older than many people think. Electric vehicles existed in the 1800s, and by 1900 a significant fraction of cars in the United States were electric. Petrol cars later dominated because of cheap fuel and long range. The modern EV revival began with the Toyota Prius hybrid in 1997 and the Tesla Roadster in 2008, followed by mass-market models such as the Nissan Leaf and the Tesla Model 3.

How an Electric Vehicle Works

An EV has four main parts: the battery pack, the electric motor, the power electronics (inverter and controller) and the charger. The battery stores direct current. The inverter converts it to alternating current for the motor, which turns the wheels. When the driver presses the accelerator, the controller sends more power to the motor. EV motors give high torque instantly, which is why EVs accelerate quickly and smoothly, and a single-speed transmission is usually enough. Electric motors are about 85 to 90 percent efficient, whereas petrol engines convert only about 25 to 35 percent of fuel energy into motion.

What Type of Battery Do Most EVs Use?

Most EVs use lithium-ion batteries. Lithium-ion cells move lithium ions between a cathode and an anode through an electrolyte during charging and discharging. They are popular because they have high energy density, long life and falling cost. Two common chemistries are NMC (nickel manganese cobalt), which offers higher range, and LFP (lithium iron phosphate), which is cheaper, safer and longer-lasting but slightly lower in energy density. The 2019 Nobel Prize in Chemistry was awarded to John Goodenough, Stanley Whittingham and Akira Yoshino for developing lithium-ion batteries. Battery capacity is measured in kilowatt-hours (kWh); a small electric scooter may have 2 to 4 kWh, a compact car 30 to 45 kWh and a long-range car 60 to 100 kWh. Batteries are managed by a Battery Management System (BMS) that monitors temperature, voltage and safety. Future technologies include solid-state batteries and sodium-ion batteries.

What Is Regenerative Braking?

Regenerative braking is a system in which the electric motor works in reverse as a generator when the driver slows down. Instead of wasting the car's kinetic energy as heat in the brake pads, the motor converts it into electricity and sends it back to charge the battery. This recovers a share of energy that would otherwise be lost, extends driving range, particularly in city stop-and-go traffic, and reduces wear on mechanical brakes. Regenerative braking does not recover all the energy and cannot fully replace friction brakes, which are still needed for hard stops.

Charging

Slow AC charging at home takes several hours. DC fast charging at public stations can add a large share of charge in 30 to 60 minutes. Common Indian connectors include CCS2 for cars and the Bharat charging standards for small vehicles. Battery swapping, where a depleted battery is exchanged for a charged one, suits two-wheelers and three-wheelers.

EVs in India

India promotes EVs through the FAME (Faster Adoption and Manufacturing of Electric Vehicles) scheme, which began in 2015, and the PM E-DRIVE scheme launched in 2024. Tamil Nadu is a major EV manufacturing hub: Ola Electric's Futurefactory is in Krishnagiri, Ather Energy has a plant in Hosur, and TVS Motor and Hyundai also produce EVs in the state. Tata Motors, Mahindra and MG offer electric cars, and electric buses run in Chennai and other cities.

Two Challenges for EV Adoption in India

Two key challenges are limited public charging infrastructure and high upfront cost. Charging stations are still few, especially outside big cities and highways, causing range anxiety, which is the fear of running out of charge. EVs also cost more than comparable petrol vehicles because the battery is expensive, and India imports most lithium-ion cells and raw materials. Other challenges include long charging times, battery recycling, dependence on a largely coal-based grid, and limited resale value certainty.

Future

The future includes cheaper batteries, solid-state technology, vehicle-to-grid systems, domestic cell manufacturing under production-linked incentives, and electrified public transport.

Quick Facts

- Most EVs use lithium-ion batteries, commonly NMC or LFP chemistry.
- Regenerative braking turns the motor into a generator to recharge the battery while slowing down.
- Two Indian EV adoption challenges are scarce charging infrastructure and high upfront cost.
- The 2019 Nobel Prize in Chemistry honoured the inventors of lithium-ion batteries.
- FAME scheme began in 2015; Tamil Nadu hosts Ola Electric in Krishnagiri and Ather in Hosur.